

Exercise 1

SQL Fundamentals : Selecting and Filtering

Q1. Retrieve all columns from employees

SELECT *

FROM employees;

ID	First Name	Last Name	Department	Salary	Hire date	City
1	Alice	Green	IT	7000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2. Unique departments

SELECT DISTINCT department

FROM employees;

Department
IT
HR
Finance
Marketing

Q3. First and Last names ordered by Salary descending

SELECT First-name, Last-name

FROM employees

ORDER BY Salary DESC;

First Name	Last Name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4. Top 3 highest-paid employees

```
SELECT id, first-name, last-name, salary
FROM employees
ORDER BY salary Desc
Limit 3;
```

id	First-name	Last-name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Q5. Employees in the IT department

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';
```

id	First-name	Last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6. Finance employees with salary > 60 000

```
SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance'
AND salary > 60000;
```

ID	First-Name	Last-Name	department	Salary
3	Cathy	Zulu	Finance	65000

Q7 Employees in HR or Marketing

```
SELECT id, First-name, Last-name, department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';
```

id	First-name	Last-name	Department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8. Employees NOT in IT

```
SELECT id, First-name, Last-name, department
FROM employees
WHERE department <> 'IT';
```

ID	First Name	Last-Name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9. Employees in IT, HR, or Finance using IN

```
SELECT id, First-name, Last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

ID	First-name	Last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q10. Employees in IT with salary > 65 000 AND city Johannesburg

```
SELECT id, First-name, Last-name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65 000
AND city = 'Johannesburg';
```

ID	First Name	Last Name	Department	Salary	City
5	Eva	Naidoo	IT	72 000	Johannesburg